

In [ ]: *#RAT-ROW*

```
In [1]: def ratrow():
        name='danish'
        for i in range(len(name)):
            for j in range(i+1):
                print(name[i],end='')
            print()
        ratrow()
```

```
d
aa
nnn
iiii
sssss
hhhhh
```

In [ ]: *#RAT-COL*

```
In [2]: def ratcol():
        name='danish'
        for i in range(len(name)):
            for j in range(i+1):
                print(name[j],end='')
            print()
        ratcol()
```

```
d
da
dan
dani
danis
danish
```

In [ ]: *#LAT-ROW*

```
In [5]: def latrow():
        name='danish'
        for i in range(len(name)):
            for j in range(len(name)-1,i,-1):
                print(' ',end='')
            for k in range(0,i+1):
                print(name[i],end='')
            print()
        latrow()
```

```
      d
      aa
      nnn
      iiii
      sssss
      hhhhhh
```

In [ ]: *#LAT-COL*

```
In [12]: def latcol():
name='danish'
for i in range(len(name)):
    for j in range(len(name)-1,i,-1):
        print(' ',end='')
    for k in range(0,i+1):
        print(name[k],end='')
    print()
latcol()
```

```

    d
   da
  dan
 dani
danis
danish
```

```
In [ ]: #PRT-ROW
```

```
In [18]: def prtrow():
name='danish'
for i in range(len(name)):
    for j in range(len(name)-1,i,-1):
        print(' ',end='')
    for k in range(0,i+1):
        print(name[i],end=' ')
    print()
prtrow()
```

```

    d
   a a
  n n n
 i i i i
s s s s s
h h h h h h
```

```
In [ ]: #PRT-COL
```

```
In [20]: def prtrow():
name='danish'
for i in range(len(name)):
    for j in range(len(name)-1,i,-1):
        print(' ',end='')
    for k in range(0,i+1):
        print(name[k],end=' ')
    print()
prtrow()
```

```

    d
   d a
  d a n
 d a n i
d a n i s
d a n i s h
```

```
In [ ]: #iratcol-upper
```

```
In [39]: def iratupper():  
    name='danish'.upper()  
    for i in range(5,-1,-1):  
        for j in range(5,i,-1):  
            print('',end='')  
        for k in range(0,i+1):  
            print(name[k],end=' ')  
        print()  
    iratupper()
```

```
D A N I S H  
D A N I S  
D A N I  
D A N  
D A  
D
```

```
In [ ]: #iratrow-upper
```

```
In [40]: def iratupper():  
    name='danish'.upper()  
    for i in range(5,-1,-1):  
        for j in range(5,i,-1):  
            print('',end='')  
        for k in range(0,i+1):  
            print(name[i],end=' ')  
        print()  
    iratupper()
```

```
H H H H H H  
S S S S S  
I I I I  
N N N  
A A  
D
```

```
In [ ]: #ilatrow-lower
```

```
In [50]: def ilatlower():  
    name='danish'.lower()  
    for i in range(5,-1,-1):  
        for j in range(5,i,-1):  
            print(' ',end=' ')  
        for k in range(0,i+1):  
            print(name[i],end=' ')  
        print()  
    ilatlower()
```

```

h h h h h h
s s s s s
i i i i
n n n
a a
d

```

In [ ]: *#ilatcol-lower*

```

In [51]: def ilatlower():
          name='danish'.lower()
          for i in range(5,-1,-1):
              for j in range(5,i,-1):
                  print(' ',end=' ')
              for k in range(0,i+1):
                  print(name[k],end=' ')
              print()
          ilatlower()

```

```

d a n i s h
d a n i s
d a n i
d a n
d a
d

```

In [ ]: *#iprtcol-upper*

```

In [45]: def iratupper():
          name='danish'.upper()
          for i in range(5,-1,-1):
              for j in range(5,i,-1):
                  print(' ',end='')
              for k in range(0,i+1):
                  print(name[k],end=' ')
              print()
          iratupper()

```

```

D A N I S H
D A N I S
D A N I
D A N
D A
D

```

In [ ]: *#iprtrow-upper*

```

In [46]: def iratupper():
          name='danish'.upper()
          for i in range(5,-1,-1):
              for j in range(5,i,-1):
                  print(' ',end='')
              for k in range(0,i+1):
                  print(name[i],end=' ')
              print()
          iratupper()

```

```

H H H H H
S S S S S
I I I I
N N N
A A
D

```

```
In [ ]: #FACTOR OF 2 DIGIT NUMBER
```

```
In [52]: def factor():
          num=int(input('ENTER THE NUMBER TO FIND THE FACTOR:'))
          for i in range(1,num+1):
              if num%i==0:
                  print(i)
          factor()

1
5
13
65
```

```
In [54]: def factor():
          num=int(input('ENTER THE NUMBER TO FIND THE FACTOR:'))
          for i in range(1,num+1):
              if num%i==0:
                  print(i)
          factor()

1
2
43
86
```

```
In [ ]: #BIGGEST NUMBER USING LAMBDA
```

```
In [60]: big = lambda a,b,c,d: a if a>b and a>c and a>d else b if b>c and b>d else c if c>d
          big(500,654,125,786)
```

```
Out[60]: 786
```

```
In [ ]: #square List using Lambda map functions
```

```
In [61]: num=[1,2,3,4,5,6,7,8,9]
          cubes = list(map(lambda a: a**2, num))
          print(cubes)

[1, 4, 9, 16, 25, 36, 49, 64, 81]
```

```
In [ ]: #cube List using Lambda map functions
```

```
In [62]: num=[4,8,12,21,36]
          cubes = list(map(lambda a: a**3, num))
          print(cubes)

[64, 512, 1728, 9261, 46656]
```